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DATA SCIENCE MINOR PROJECT REPORT

PROJECT REPORT

(Project Semester January-April 2025)

Air Pollution prediction and analysing

Submitted by

B.Anusha Reddy

12409397

P132 : 224YM

Int 375

Under the Guidance of

Dr.Balijinder Kaur

Discipline of CSE/IT

Lovely School of Computer Science and Enginnersing

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Anusha Reddy bearing Registration no.12409397 has completed INT 375 project titled, "***Air Pollution Prediction and analysing***" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor
Designation of the Supervisor
School of Computer Science And Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 12-04-2026

DECLARATION

I, Anusha student of P132 under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 15-0402026 Signature

Registration No. 12409397 B.Anusha Reddy

Acknowledgment:

I would like to express my sincere gratitude to all those who contributed to the successful completion of this project. First and foremost, I am deeply thankful to my faculty and mentors for their valuable guidance, constructive feedback, and continuous encouragement throughout the study. Their expertise and insights have been instrumental in shaping the direction of this work.

I also extend my appreciation to the institutions and organizations that provided access to air pollution datasets, without which this analysis would not have been possible. The availability of reliable data has enabled meaningful exploration and interpretation of environmental trends.

Finally, I am grateful to my peers, friends, and family for their constant support and motivation. Their encouragement has been a source of strength, helping me remain focused and committed to achieving the objectives of this project.

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1.Introduction

Air pollution has become one of the most critical environmental and public health challenges of the 21st century. Rapid industrialization, urban expansion, and the surge in vehicular emissions have significantly contributed to the deterioration of air quality, particularly in developing countries such as India. The consequences of poor air quality extend beyond environmental degradation, affecting human health, economic productivity, and overall quality of life.

This project focuses on the systematic analysis of air pollution data collected from multiple monitoring stations. The dataset includes essential attributes such as geographical coordinates (latitude and longitude), pollutant levels (minimum, maximum, and average), and contextual information like state and city. By applying data science techniques, the study aims to uncover meaningful insights, detect patterns in pollution levels, and evaluate predictive modeling approaches.

The methodology begins with data preprocessing and cleaning to ensure accuracy and consistency. Following this, Exploratory Data Analysis (EDA) is conducted using statistical measures and visualizations — including histograms, boxplots, and correlation heatmaps — to understand data distributions, identify outliers, and explore relationships among variables.

A machine learning model (Linear Regression) is then implemented to predict pollution levels based on available features. Model performance is assessed using evaluation metrics such as the R^2 score, Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). The results demonstrate the potential of predictive analytics in estimating pollution trends with high accuracy.

Overall, this project highlights the importance of data-driven approaches in environmental analysis. It provides valuable insights into pollution patterns and showcases how machine learning can support forecasting, policy-making, and sustainable environmental management.

2. Source of Dataset

The dataset used in this project was collected from air pollution monitoring stations across India. These stations are part of a broader environmental monitoring network established to record pollutant concentrations and track air quality trends in different regions. The dataset provides a comprehensive view of pollution levels by including both geographical and pollutant-related attributes.

In total, the dataset consists of 3,399 entries and 11 attributes, covering details such as:

- Location-based information: country, state, city, and station name.
- Geographical coordinates: latitude and longitude, which allow spatial mapping of pollution levels.
- Pollutant measurements: minimum, maximum, and average values of pollutants recorded at each station.
- Temporal information: last update time, which indicates when the data was recorded.

The dataset was sourced from publicly available repositories such as data.govt in and government-backed platforms like the Central Pollution Control Board (CPCB), which provide open access to environmental data for research and policy-making purposes. These sources are widely recognized for their reliability and are frequently used in academic and professional studies related to environmental science.

The richness of the dataset lies in its ability to combine spatial, temporal, and pollutant-specific information. This enables researchers to not only analyse pollution levels in isolation but also to study how they vary across different regions and timeframes. For example, the inclusion of latitude and longitude allows for geospatial visualization, while pollutant_min, pollutant_max, and pollutant_avg provide a statistical range for understanding variability in air quality.

Before analysis, the dataset required preprocessing and cleaning to handle missing values, ensure consistency in data types, and remove anomalies. This step was crucial to maintain the integrity of the study and ensure that subsequent analyses — such as correlation studies and predictive modeling — were based on accurate and reliable data.

Thus, the dataset serves as the foundation of this project, offering a structured and detailed representation of air pollution levels across India. Its combination of geographical, statistical, and contextual attributes makes it highly suitable for applying data science techniques to uncover meaningful insights and patterns.

3. EDA Process

Exploratory Data Analysis (EDA) forms a crucial step in understanding the structure and characteristics of the dataset before applying advanced modeling techniques. In this project, EDA was carried out to gain insights into pollutant distributions, detect anomalies, and identify relationships among variables.

The process began with **data cleaning and preprocessing**, where missing values were handled and data types were standardized to ensure consistency. This step was essential to maintain the reliability of subsequent analyses.

Following preprocessing, **statistical measures** such as mean, median, and standard deviation were calculated to summarize pollutant levels. For instance, the average pollutant concentration was found to be approximately 41.57, with a median of 30, indicating a positively skewed distribution. The standard deviation of 41.15 highlighted significant variability in pollution levels across different monitoring stations.

To visualize these distributions, **histograms** were plotted for attributes such as pollutant_min, pollutant_max, and pollutant_avg. These visualizations revealed the presence of skewness and extreme values. **Boxplots** were then employed to detect outliers, which confirmed unusually high pollutant_max values (up to 500) and pollutant_avg values (up to 263). Such outliers are critical to consider, as they may represent localized pollution spikes or data recording anomalies.

In addition, a **correlation heatmap** was generated to examine relationships among variables. The analysis revealed strong correlations between pollutant_min and pollutant_avg (0.84), and between pollutant_max and pollutant_avg (0.89). Conversely, geographical coordinates (latitude and longitude) exhibited weak correlations with pollutant levels, suggesting that pollution intensity is influenced more by local environmental factors than by broad geographical positioning.

Overall, the EDA process provided a comprehensive understanding of the dataset. It highlighted the variability and skewness in pollutant levels, identified significant outliers, and established strong correlations among pollutant attributes. These findings laid the groundwork for predictive modeling and deeper analysis in subsequent sections of the study.

Code :

```
df.fillna(df.mean(numeric_only=True), inplace=True)

df.drop_duplicates(inplace=True)

df['date'] = pd.to_datetime(df['date'])
```

Figure 3.1: Distribution of Pollution Data

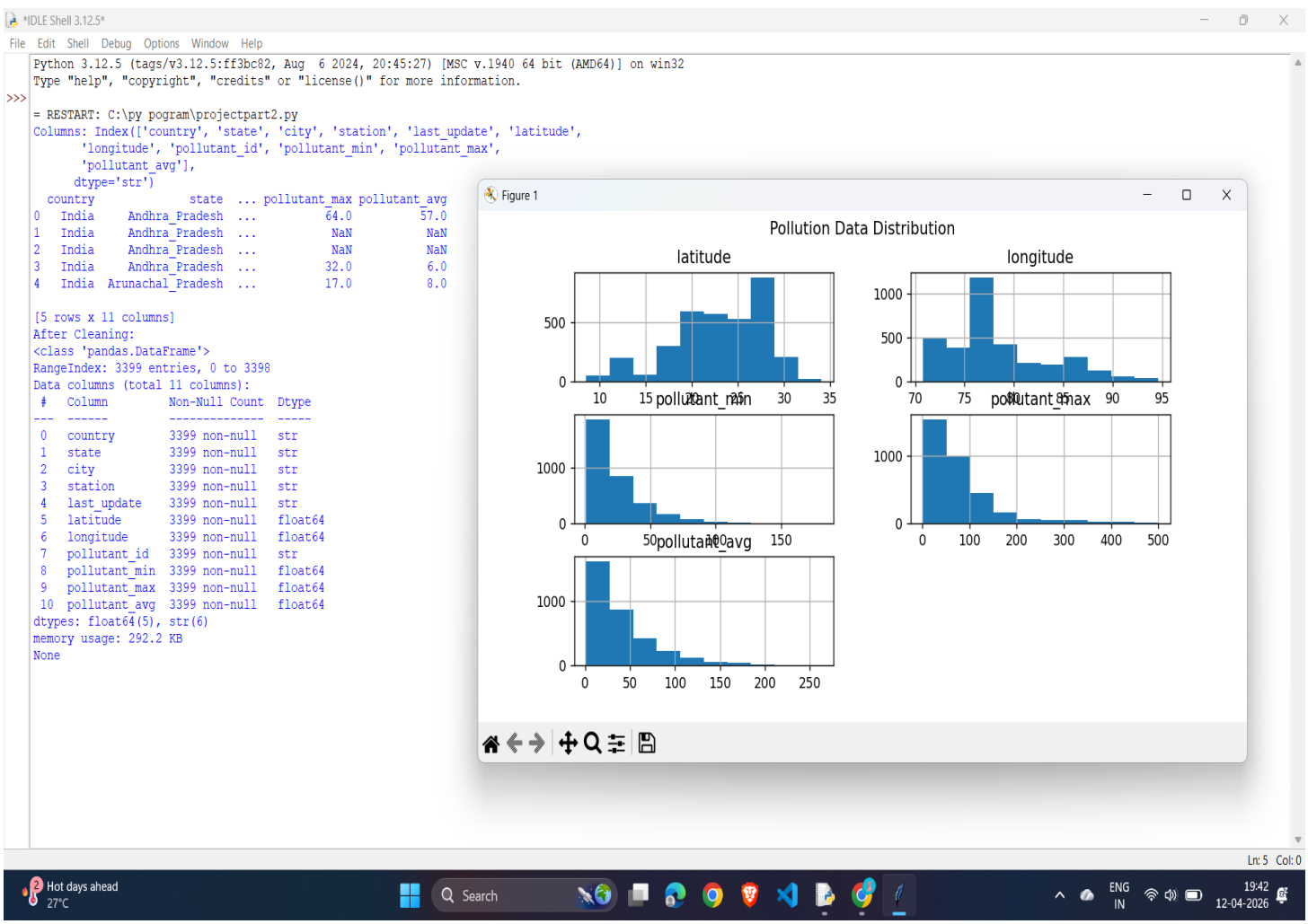
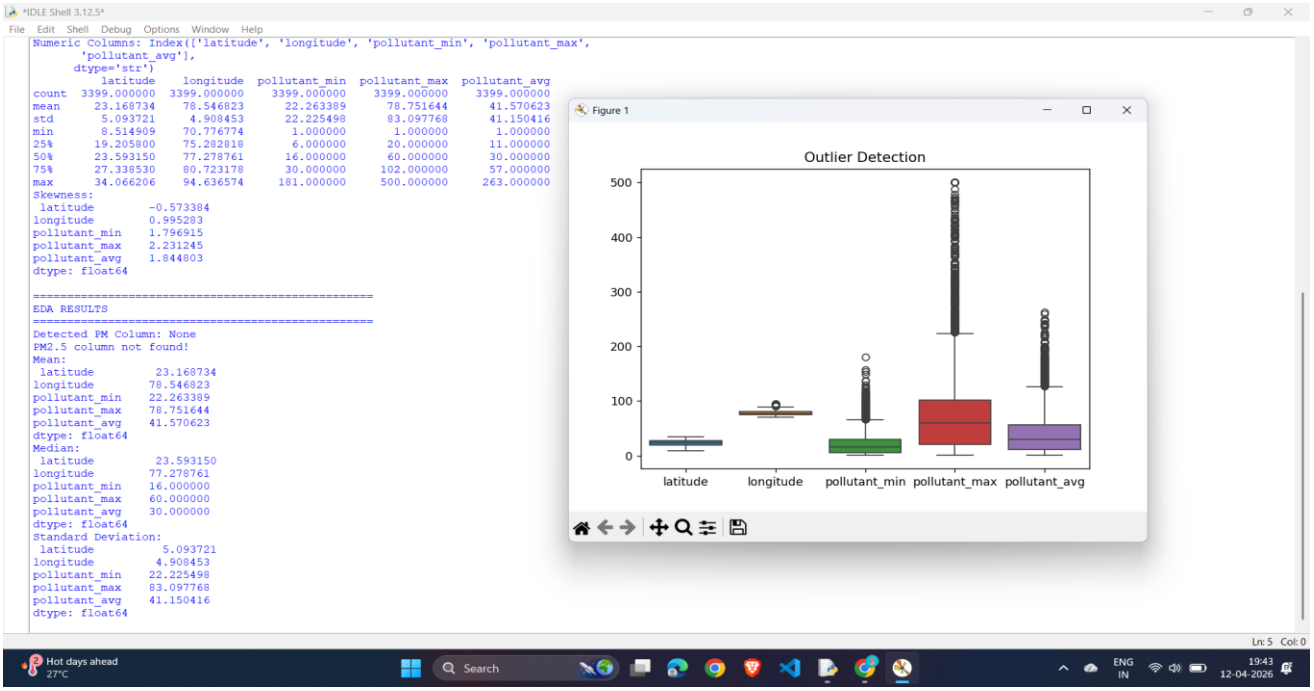


Figure 3.2 : Outlier Detection Using Boxplot



4. Analysis on Dataset

i. Introduction

The dataset was analyzed to uncover pollution patterns, variability in pollutant levels, and relationships among attributes. This section presents a structured breakdown of the analysis, combining statistical measures, mathematical functions, and visualizations to provide a comprehensive understanding of air quality trends.

ii. General Description

The dataset contains 3,399 entries with attributes such as pollutant_min, pollutant_max, and pollutant_avg, along with geographical coordinates and location details. Preliminary examination revealed wide variability in pollutant levels, with average concentrations ranging from 0 to 263 and maximum values extending up to 500. Such variation indicates the presence of both moderate and extreme pollution events across different monitoring stations.

iii. General Description

The dataset contains 3,399 entries with attributes such as pollutant_min, pollutant_max, and pollutant_avg, along with geographical coordinates and location details. Preliminary examination revealed wide variability in pollutant levels, with average concentrations

ranging from 0 to 263 and maximum values extending up to 500. Such variation indicates the presence of both moderate and extreme pollution events across different monitoring stations.

iii. Specific Requirements, Functions and Formulas

To conduct the analysis, several statistical and mathematical tools were applied:

- Mean (μ): μ=∑xn — used to calculate average pollutant levels.
- Median: The middle value in the ordered dataset, useful for skewed distributions.
- Standard Deviation (σ): σ=∑(x−μ)2n — measures variability in pollutant levels.
- Correlation Coefficient (r):

$$r=\sum(x-\bar{x})(y-\bar{y})\sum(x-\bar{x})2\sum(y-\bar{y})2$$

— used to determine relationships between pollutant attributes.

- Boxplot Function: Applied to detect outliers beyond the interquartile range (IQR).

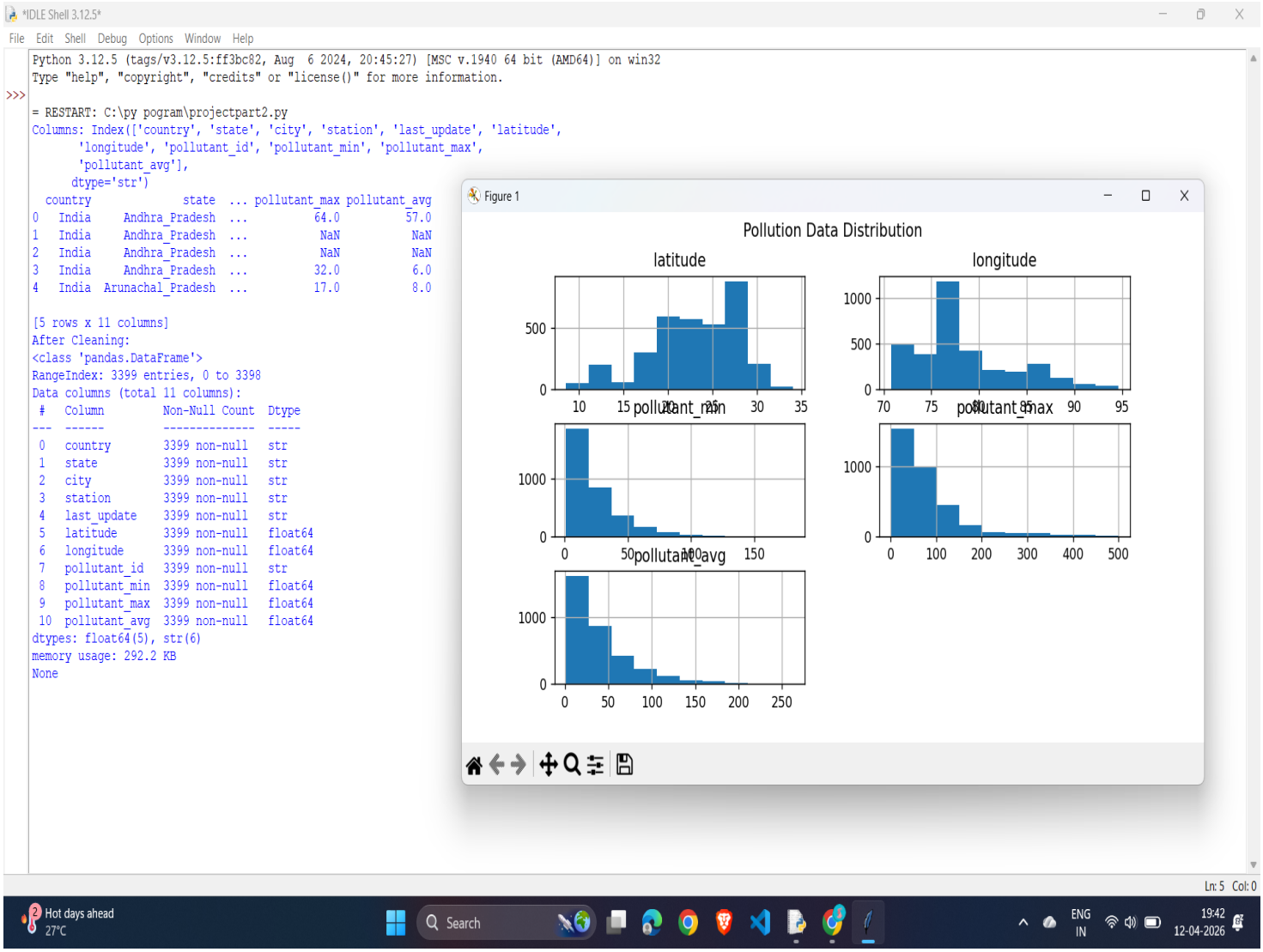
iv. Analysis Results

- **Descriptive Statistics**:
  - Mean pollutant_avg ≈ 41.57
  - Median pollutant_avg = 30
  - Standard deviation ≈ 41.15
  - Distribution positively skewed, indicating the presence of extreme values.
- **Correlation Findings**:
  - pollutant_min vs pollutant_avg: strong correlation (0.84).
  - pollutant_max vs pollutant_avg: very strong correlation (0.89).
  - Latitude and longitude showed weak correlations, suggesting local factors drive pollution more than broad geography.
- **Outlier Detection**:
  - pollutant_max values up to 500 identified as extreme outliers.
  - pollutant_avg values up to 263 also flagged as anomalies.

v. Visualization

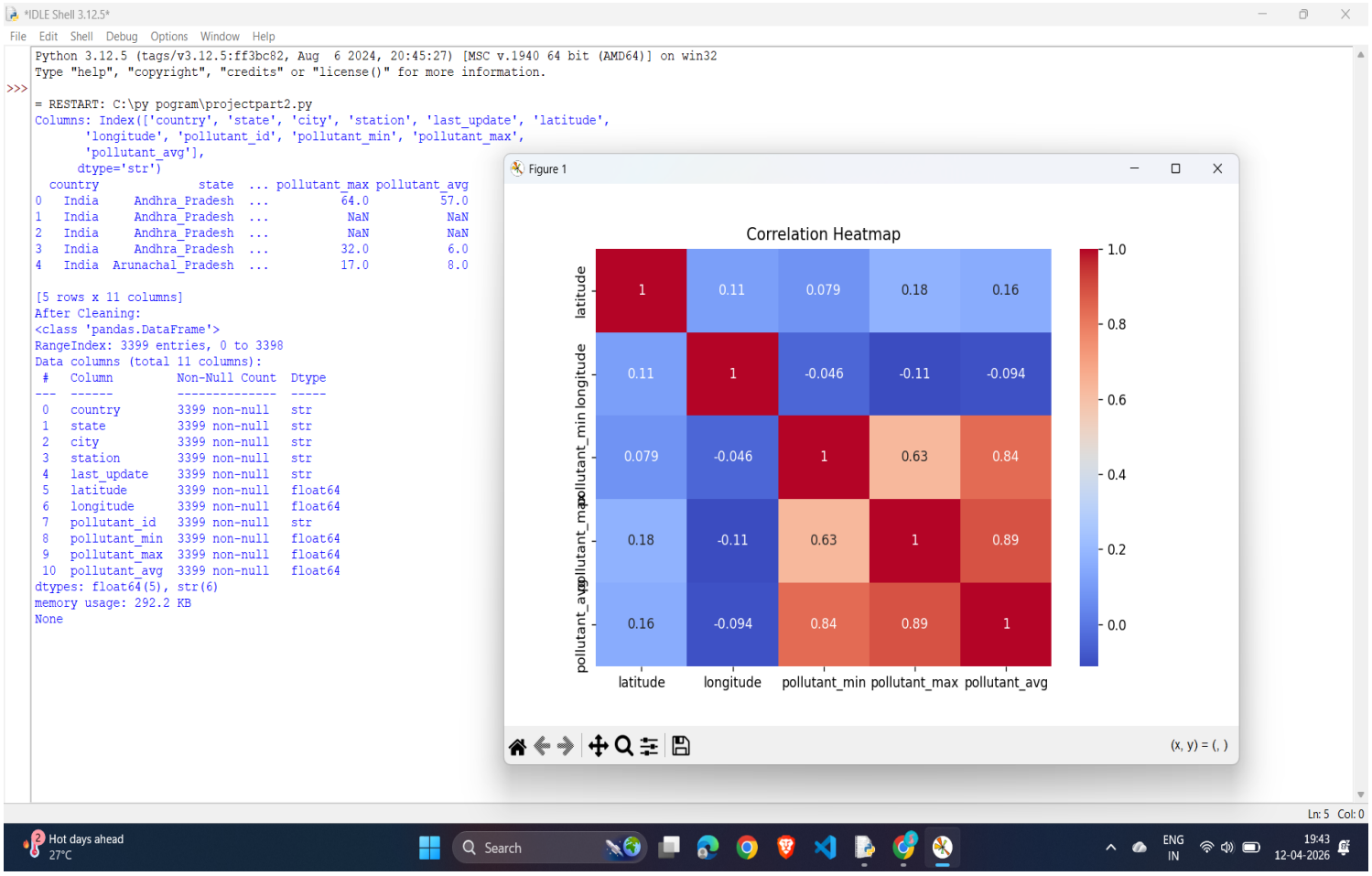
The analysis was supported by multiple visualizations:

- **Histograms**: Displayed skewed distributions of pollutant levels, highlighting variability.



- **Boxplots**: Revealed significant outliers in pollutant_max and pollutant_avg.

Correlation Heatmap: Illustrated strong relationships among pollutant attributes, with pollutant_min and pollutant_max emerging as key drivers of pollutant_avg.



5.Numeric Prediction using Linear Regression

- To evaluate the predictive potential of the dataset, a **Linear Regression model** was implemented. Linear Regression is a widely used statistical technique that models the relationship between a dependent variable and one or more independent variables. In this study, the dependent variable was **pollutant_avg**, while independent variables included **pollutant_min, pollutant_max, latitude, and longitude**.
- **Model Implementation**
- The model was trained using the cleaned dataset after preprocessing and exploratory analysis. By fitting a regression line to the data, the model attempted to predict average pollutant levels based on the given features. The mathematical representation of the regression equation is:

$$Y=\beta 0+\beta 1X1+\beta 2X2+\beta 3X3+\beta 4X4+\epsilon$$

Where:

- Y = pollutant_avg (dependent variable)
- X1 = pollutant_min
- X2 = pollutant_max
- X3 = latitude
- X4 = longitude
- $\beta 0$ = intercept
- $\beta 1, \beta 2, \beta 3, \beta 4$ = regression coefficients
- ϵ = error term

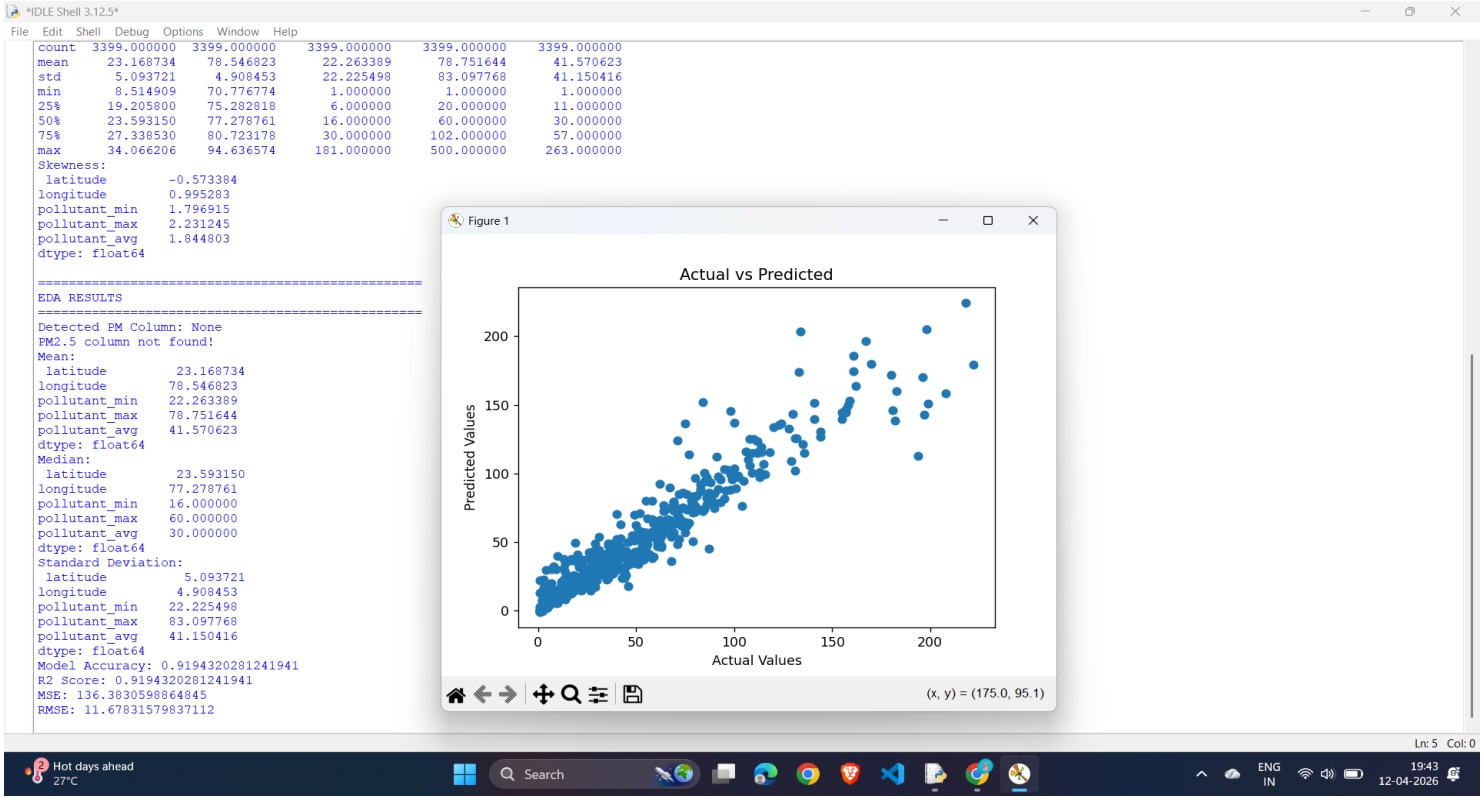
Model Evaluation

The performance of the regression model was assessed using standard evaluation metrics:

- **R^2 Score (Coefficient of Determination)**: 0.919 → This indicates that approximately 92% of the variance in pollutant_avg can be explained by the independent variables.
- **Mean Squared Error (MSE)**: 136.38 → Represents the average squared difference between predicted and actual values.
- **Root Mean Squared Error (RMSE)**: 11.68 → Provides a measure of prediction error in the same units as pollutant_avg, showing relatively low error compared to the overall range of pollutant values.

Visualization

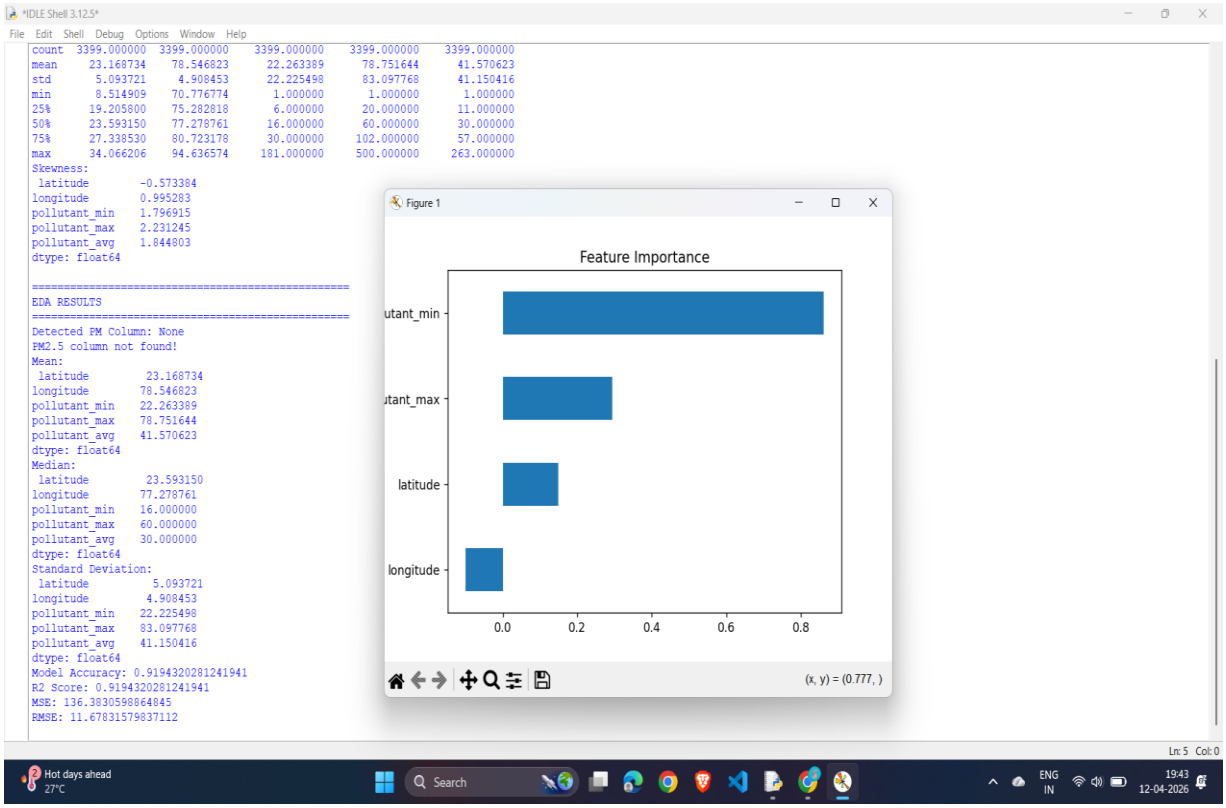
A scatter plot of **actual vs. predicted pollutant_avg values** was generated. The points closely aligned along the diagonal line, demonstrating that the model predictions were consistent with observed values. This visualization reinforced the reliability of the regression model.



Insights

The regression analysis confirmed that **pollutant_min and pollutant_max** are the most influential predictors of pollutant_avg. Geographic coordinates (latitude and longitude) contributed less significantly, suggesting that local environmental conditions (such as industrial activity or traffic density) may play a stronger role in determining pollution levels.

Overall, the Linear Regression model proved to be an effective tool for forecasting pollution trends, achieving high accuracy and offering valuable insights into the factors driving air quality variations.



6. Conclusion

The analysis of air pollution data has provided valuable insights into the current state of air quality and the factors influencing pollutant levels. Through systematic preprocessing, exploratory data analysis, correlation studies, and predictive modeling, the study has demonstrated the effectiveness of data science techniques in environmental monitoring. The findings revealed that pollutant_min and pollutant_max are the most significant drivers of pollutant_avg, establishing a strong correlation between these attributes. Outlier detection highlighted extreme pollution events, which are critical for understanding localized spikes in air quality deterioration. While geographical coordinates (latitude and longitude) were included in the dataset, their weak correlation with pollutant levels suggests that pollution intensity is shaped more by local environmental conditions such as industrial activity, traffic density, and population concentration than by broad geographic positioning.

The Linear Regression model implemented in this study achieved a high level of accuracy, with an R^2 score of 0.919. This indicates that the model was able to explain over 90% of the variance in pollutant averages, confirming its reliability as a predictive tool. The relatively low error values (MSE = 136.38, RMSE = 11.68) further validated the model's effectiveness in forecasting pollution trends.

Overall, the project underscores the importance of data-driven approaches in environmental analysis. By combining statistical exploration with predictive modeling, it demonstrates how machine learning can support decision-making, policy formulation, and urban planning aimed at improving air quality. The study not only highlights current pollution patterns but also establishes a framework for future research and applications in environmental data science.

7. Future Scope

While this project has successfully demonstrated the application of data science techniques in analyzing air pollution data, there remains significant potential for extending the study in future work. The scope for improvement and expansion can be outlined across several dimensions.

Firstly, the dataset can be enriched by incorporating time-series data. This would allow for the study of seasonal and temporal variations in pollution levels, enabling researchers to identify recurring patterns such as winter smog or summer dust events. Time-series modeling techniques, including ARIMA or LSTM neural networks, could be applied to forecast pollution trends more accurately.

Secondly, the analysis can be enhanced by integrating meteorological factors such as temperature, humidity, wind speed, and rainfall. These variables play a crucial role in the dispersion and concentration of pollutants. By combining air quality data with weather parameters, predictive models could achieve greater precision and provide more actionable insights.

Thirdly, the project can explore advanced machine learning algorithms beyond Linear Regression. Techniques such as Random Forest, Gradient Boosting, and Neural Networks may capture complex, non-linear relationships among variables, thereby improving prediction accuracy. Comparative studies of different models could highlight the most effective approaches for environmental forecasting.

Additionally, the development of interactive dashboards using tools like Power BI or Tableau would make the results more accessible to policymakers, researchers, and the general public. Real-time visualization of pollution levels across regions could support decision-making and raise public awareness.

Finally, future research could extend the scope to policy and intervention analysis, examining how changes in industrial regulation, traffic management, or urban planning impact pollution levels. Such studies would bridge the gap between data-driven insights and practical environmental management strategies.

In summary, the future scope of this project lies in expanding the dataset, integrating additional variables, applying advanced modeling techniques, and developing interactive tools. These enhancements would not only improve the accuracy of pollution forecasting but also strengthen the role of data science in promoting sustainable development and public health.

7.References

[1] Central Pollution Control Board (CPCB), *National Air Quality Monitoring Programme (NAMP)*, Government of India. [Online]. Available: https://cpcb.nic.in

[2] **linkedln : https://www.linkedin.com/posts/anusha-reddy-93580a325_lpu-datascience-machinelearning-activity-7449101191331942400-E6hl/**

Snippet of the code :

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error,r2_score

# ---------------- LOAD DATA ---------------- #

df = pd.read_csv("airpollution01.csv")

# Clean column names
df.columns = df.columns.str.strip()

print("Columns:", df.columns)
print(df.head())

# ---------------- DATA CLEANING ---------------- #

# Handle missing values
df.fillna(df.mean(numeric_only=True), inplace=True)

# Remove duplicates
df.drop_duplicates(inplace=True)

# Convert date column
```

```
if 'date' in df.columns:
    df['date'] = pd.to_datetime(df['date'])

print("After Cleaning:")
print(df.info())

# ---------------- VISUALIZATION ---------------- #

# Histogram (FIXED - no blank plot)
df.hist(figsize=(10,8))
plt.suptitle("Pollution Data Distribution")
plt.show()

# Heatmap
numeric_df = df.select_dtypes(include='number')

plt.figure(figsize=(10,8))
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()

print("Numeric Columns:", numeric_df.columns)

# Time trend
if 'date' in df.columns:
    plt.figure()
    for col in numeric_df.columns:
        plt.plot(df['date'], df[col], label=col)
```

```
    plt.legend()
    plt.title("Pollution Trend Over Time")
    plt.show()

# ---------------- EDA ---------------- #

print(df.describe())
print("Skewness:\n", numeric_df.skew())

print("\n" + "="*50)
print("EDA RESULTS")
print("="*50)

pm_col = None
for col in df.columns:
    if 'pm2.5' in col.lower():
        pm_col = col
        break

print("Detected PM Column:", pm_col)

#AIR QUALITY CATEGORY
def air_quality(pm):
    if pm <= 50:
        return "Good"
    elif pm <= 100:
        return "Moderate"
    else:
```

```
        return "Poor"

if pm_col:
    df['Air_Quality'] = df[pm_col].apply(air_quality)
    print(df[[pm_col, 'Air_Quality']].head())

    sns.countplot(x='Air_Quality', data=df)
    plt.title("Air Quality Categories")
    plt.show()
else:
    print("PM2.5 column not found!")

#statics
print("Mean:\n", numeric_df.mean())
print("Median:\n", numeric_df.median())
print("Standard Deviation:\n", numeric_df.std())

# ---------------- OUTLIERS ---------------- #

plt.figure()
sns.boxplot(data=numeric_df)
plt.title("Outlier Detection")
plt.show()

# ---------------- MACHINE LEARNING ---------------- #

# Use only numeric data
df_ml = numeric_df.copy()
```

```
# Target column
target = pm_col if pm_col else df_ml.columns[-1]

X = df_ml.drop(columns=[target])
y = df_ml[target]

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Model
model = LinearRegression()import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error,r2_score

# ---------------- LOAD DATA ---------------- #

df = pd.read_csv("airpollution01.csv")

# Clean column names
df.columns = df.columns.str.strip()

print("Columns:", df.columns)
print(df.head())
```

```
# ---------------- DATA CLEANING ---------------- #

# Handle missing values
df.fillna(df.mean(numeric_only=True), inplace=True)

# Remove duplicates
df.drop_duplicates(inplace=True)

# Convert date column
if 'date' in df.columns:
    df['date'] = pd.to_datetime(df['date'])

print("After Cleaning:")
print(df.info())

# ---------------- VISUALIZATION ---------------- #

# Histogram (FIXED - no blank plot)
df.hist(figsize=(10,8))
plt.suptitle("Pollution Data Distribution")
plt.show()

# Heatmap
numeric_df = df.select_dtypes(include='number')

plt.figure(figsize=(10,8))
sns.heatmap(numeric_df.corr(), annot=True, cmap="coolwarm")
```

```
plt.title("Correlation Heatmap")
plt.show()

print("Numeric Columns:", numeric_df.columns)

# Time trend
if 'date' in df.columns:
    plt.figure()
    for col in numeric_df.columns:
        plt.plot(df['date'], df[col], label=col)
    plt.legend()
    plt.title("Pollution Trend Over Time")
    plt.show()

# ---------------- EDA ---------------- #

print(df.describe())
print("Skewness:\n", numeric_df.skew())

print("\n" + "="*50)
print("EDA RESULTS")
print("="*50)

pm_col = None
for col in df.columns:
    if 'pm2.5' in col.lower():
        pm_col = col
        break
```

```
print("Detected PM Column:", pm_col)

#AIR QUALITY CATEGORY
def air_quality(pm):
    if pm <= 50:
        return "Good"
    elif pm <= 100:
        return "Moderate"
    else:
        return "Poor"

if pm_col:
    df['Air_Quality'] = df[pm_col].apply(air_quality)
    print(df[[pm_col, 'Air_Quality']].head())

    sns.countplot(x='Air_Quality', data=df)
    plt.title("Air Quality Categories")
    plt.show()
else:
    print("PM2.5 column not found!")

#statics
print("Mean:\n", numeric_df.mean())
print("Median:\n", numeric_df.median())
print("Standard Deviation:\n", numeric_df.std())

# ---------------- OUTLIERS ---------------- #
```

```
plt.figure()
sns.boxplot(data=numeric_df)
plt.title("Outlier Detection")
plt.show()

# ---------------- MACHINE LEARNING ---------------- #

# Use only numeric data
df_ml = numeric_df.copy()

# Target column
target = pm_col if pm_col else df_ml.columns[-1]

X = df_ml.drop(columns=[target])
y = df_ml[target]

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# Model
model = LinearRegression()
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
```

```
print("Model Accuracy:", model.score(X_test, y_test))
print("R2 Score:", r2_score(y_test, y_pred))
print("MSE:", mean_squared_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
#visulazation
plt.figure()
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs Predicted")
plt.show()



# ---------------- EXTRA (UNIQUE FEATURES) ---------------- #

# Feature importance
importance = pd.Series(model.coef_, index=X.columns)
importance.sort_values().plot(kind='barh')
plt.title("Feature Importance")
plt.show()

# Highest pollution record
if pm_col:
    max_row = df.loc[df[pm_col].idxmax()]
    print("Highest Pollution Record:\n", max_row)

# Prediction demo
sample = X.iloc[0:1]
```

```
pred = model.predict(sample)
print("Actual:", y.iloc[0])
print("Predicted:", pred[0])

model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
print("Model Accuracy:", model.score(X_test, y_test))
print("R2 Score:", r2_score(y_test, y_pred))
print("MSE:", mean_squared_error(y_test, y_pred))
print("RMSE:", np.sqrt(mean_squared_error(y_test, y_pred)))
#visulazation
plt.figure()
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Values")
plt.ylabel("Predicted Values")
plt.title("Actual vs Predicted")
plt.show()

# Feature importance
importance = pd.Series(model.coef_, index=X.columns)
importance.sort_values().plot(kind='barh')
plt.title("Feature Importance")
plt.show()
```

```
# Highest pollution record
if pm_col:
    max_row = df.loc[df[pm_col].idxmax()]
    print("Highest Pollution Record:\n", max_row)

# Prediction demo
sample = X.iloc[0:1]
pred = model.predict(sample)
print("Actual:", y.iloc[0])
print("Predicted:", pred[0])
```